Hopf algebras, Hopf monads, and quantum invariants

Alexis Virelizier

(Université Montpellier 2)

Based on joint works with:

- Alain Bruguières [B.V.]
- Alain Bruguières and Steve Lack [B.L.V.]
- Vladimir Turaev [T.V.]

Conformal field theories and tensor categories

BICMR, Beijing, 13 June 2011

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

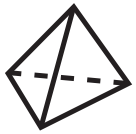
Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation  + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

▹ mathematical realization of Witten's Chern-Simons invariants

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

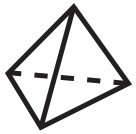
presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

▹ mathematical realization of Witten's Chern-Simons invariants

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation  + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

▹ related to the Ponzano-Regge model for 3-dim quantum gravity

Question

Compare the invariants RT and TV

Quantum invariants of 3-manifolds

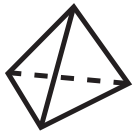
presentation of M^3 + algebraic/categorical data $\rightsquigarrow$ invariant

- **Braided setting:** *Reshetikhin-Turaev (91)*

surgery + $\mathcal{B}$ fusion modular category $\rightsquigarrow$ $\mathrm{RT}_{\mathcal{B}}(M)$

▹ mathematical realization of Witten's Chern-Simons invariants

- **Non-braided setting:** *Turaev-Viro (92)*

triangulation  + $\mathcal{C}$ fusion spherical category, $\dim \mathcal{C} \neq 0$ $\rightsquigarrow$ $\mathrm{TV}_{\mathcal{C}}(M)$

▹ related to the Ponzano-Regge model for 3-dim quantum gravity

Question

Compare the invariants RT and TV

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C})$ = *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C}) =$ *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C}) =$ *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C})$ = *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C}) =$ *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

The Drinfeld-Joyal-Street center

$\mathcal{C}$ monoidal category $\xrightarrow[\text{construction}]{\text{center}}$ $\mathcal{Z}(\mathcal{C})$ braided category

- Objects of $\mathcal{Z}(\mathcal{C}) =$ *half-braidings* of $\mathcal{C}$:
  pair (X, σ) with $\sigma = \left\{\sigma_Y \colon X \otimes Y \xrightarrow{\sim} Y \otimes X\right\}_{Y \in \mathcal{C}}$ such that
  $$\sigma_{Y \otimes Z} = (\mathrm{id}_Y \otimes \sigma_Z)(\sigma_Y \otimes \mathrm{id}_Z)$$
- $f \colon (X, \sigma) \to (X', \sigma')$ in $\mathcal{Z}(\mathcal{C})$ is $f \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ such that
  $$\sigma'(f \otimes \mathrm{id}) = (\mathrm{id} \otimes f)\sigma$$
- $(X, \sigma) \otimes_{\mathcal{Z}(\mathcal{C})} (X', \sigma') = \big(X \otimes X', (\sigma_Y \otimes \mathrm{id})(\mathrm{id} \otimes \sigma')\big)$
- Braiding: $c_{(X,\sigma),(X',\sigma')} = \sigma_{X'}$

Example : H Hopf algebra over $\Bbbk \rightsquigarrow \mathcal{Z}(H\text{-}\mathrm{mod}) \cong D(H)\text{-}\mathrm{mod}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

- **Braided case:** *Turaev (93), Roberts (95)*

  $\mathcal{B}$ fusion modular $\implies$ $\mathcal{B}$ spherical fusion and $\dim \mathcal{B} \neq 0$

$$\mathrm{TV}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M)\,\mathrm{RT}_{\mathcal{B}}(-M) \underset{\substack{\uparrow \\ \textit{(if } \mathcal{B} \textit{ hermitian)}}}{=} \left|\mathrm{RT}_{\mathcal{B}}(M)\right|^2$$

- **Non-braided case:** *Müger (02)*

  $\mathcal{C}$ fusion spherical category over $\Bbbk$ alg. closed field, $\dim \mathcal{C} \neq 0$

  $\implies$ the center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ is modular

Conjecture (Turaev and al., 02)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

▹ implies Turaev-Roberts: if $\mathcal{B}$ braided, then $\mathcal{Z}(\mathcal{B}) \simeq \mathcal{B} \boxtimes \mathcal{B}^{\mathrm{op}}$

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:

the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:

the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$

but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

The Reshetikhin-Turaev invariant (1991)

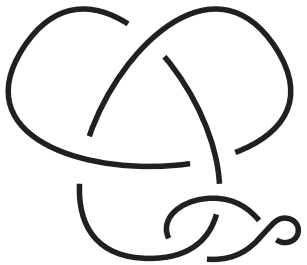
$\mathcal{C}$ spherical fusion with $\dim \mathcal{C} \neq 0 \rightsquigarrow \mathcal{Z}(\mathcal{C})$ modular category

M^3 presented by surgery along framed link $L = L_1 \cup \cdots \cup L_n$

The Reshetikhin-Turaev invariant (1991)

$\mathcal{C}$ spherical fusion with $\dim \mathcal{C} \neq 0 \rightsquigarrow \mathcal{Z}(\mathcal{C})$ modular category

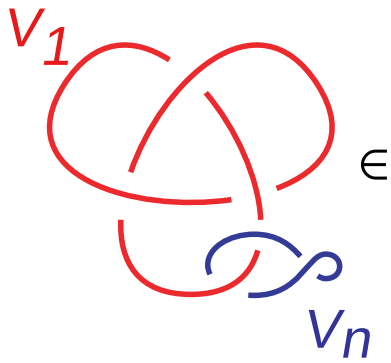
M^3 presented by surgery along framed link $L = L_1 \cup \cdots \cup L_n$



The Reshetikhin-Turaev invariant (1991)

$\mathcal{C}$ spherical fusion with $\dim \mathcal{C} \neq 0 \rightsquigarrow \mathcal{Z}(\mathcal{C})$ modular category

M^3 presented by surgery along framed link $L = L_1 \cup \cdots \cup L_n$



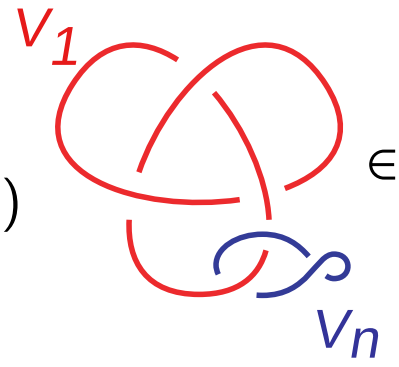
$\in \mathrm{End}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1})$
$= \Bbbk$

The Reshetikhin-Turaev invariant (1991)

$\mathcal{C}$ spherical fusion with $\dim \mathcal{C} \neq 0 \rightsquigarrow \mathcal{Z}(\mathcal{C})$ modular category

M^3 presented by surgery along framed link $L = L_1 \cup \cdots \cup L_n$

$$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \sum_{\substack{V_1,\ldots,V_n \text{ simple} \\ \text{objects of } \mathcal{Z}(\mathcal{C})}} \dim_q(V_1)\cdots\dim_q(V_n)$$



$\in \mathrm{End}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1})$
$= \Bbbk$

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:

the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$

but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:
the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:
the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:
the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:
the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim(\mathcal{C}) \neq 0$

Theorem (T.V., 2010)

$$\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$$

Difficulty:
the construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ involves the simple objects of $\mathcal{Z}(\mathcal{C})$ but no workable description of the simple objects of $\mathcal{Z}(\mathcal{C})$ is known

1. Alternative construction of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
   - Hopf algebras (in braided categories) play a central role (especially the *coend* of a ribbon category)
   - Hopf diagrams allow computations
2. Alternative description of $\mathcal{Z}(\mathcal{C})$ via Hopf monads
   - Hopf monads extend Hopf algebras to the non-braided setting
   - Construction of the double of a Hopf monad (gives a 'Hopf algebraic' description of the Drinfeld center)

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$, $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$, $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$ $\sim$

, $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
   $$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
   Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

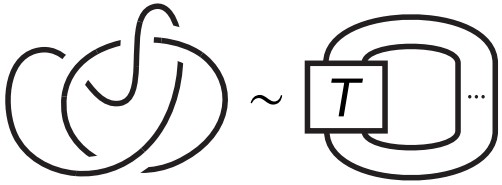
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

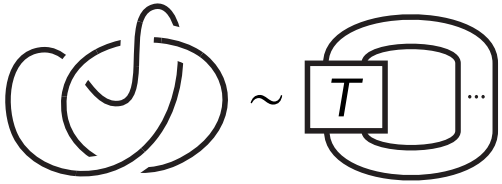
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
   $$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
   Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

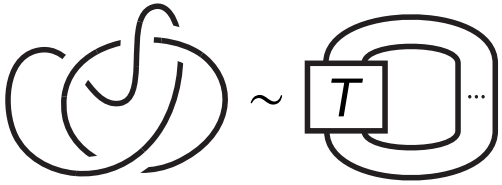
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

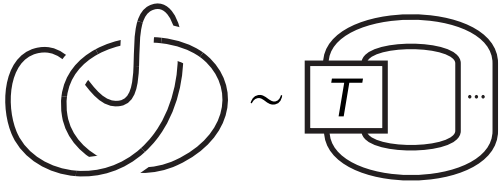
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$ 

, $T =$

$\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

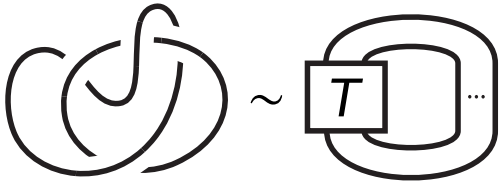
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
   $$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
   Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

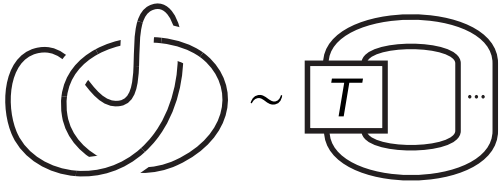
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

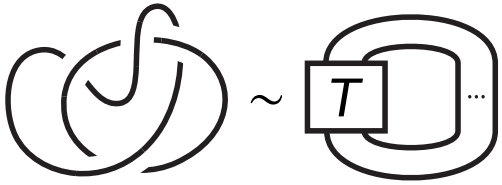
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

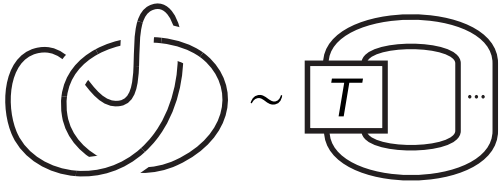
$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

A universal surgical approach

$\mathcal{B}$ ribbon category admitting a ***coend*** $C = \int^{X \in \mathcal{B}} X^* \otimes X$

$\rightsquigarrow$ C is a Hopf algebra in $\mathcal{B}$, with a Hopf pairing *(Deligne, Majid)*

$L =$  , $T =$ $\xrightarrow[\text{(Lyubashenko)}]{\text{univ. property}}$ $\phi_T \colon C^{\otimes n} \to \mathbb{1}$

Theorem (V. 2003)

1. If $\alpha \colon \mathbb{1} \to C$ is a *Kirby element*, i.e.,
$$(\mathrm{id}_C \otimes m_C)(\Delta_C \otimes \mathrm{id}_C)(\alpha \otimes \alpha) = \alpha \otimes \alpha \quad \text{and} \quad S_C(\alpha) = \alpha$$
Then $\tau_{\mathcal{B}}(S^3_L; \alpha) = \langle \phi_T, \alpha^{\otimes n} \rangle$ is an invariant of 3-manifolds
2. There exist 'non-semisimple' Kirby elements

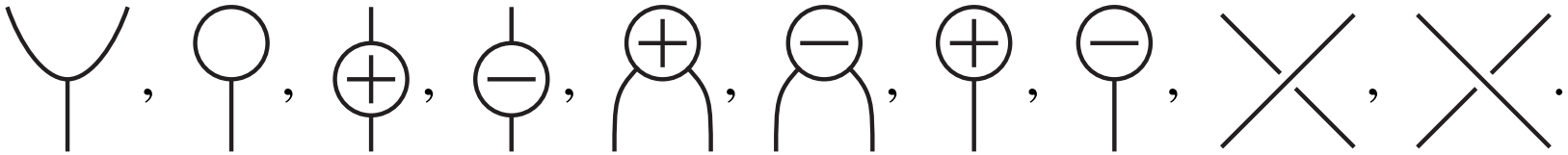
Example: $\alpha = \lambda \colon \mathbb{1} \to C$ integral of C:

$$\tau_{\mathcal{B}}(M; \lambda) = \mathrm{Lyub}_{\mathcal{B}}(M) = \mathrm{RT}_{\mathcal{B}}(M) \text{ if } \mathcal{B} \text{ is fusion}$$

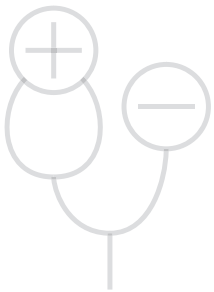
Problem: How to compute ϕ_T ? $\rightsquigarrow$ *Hopf diagrams*

Hopf diagrams

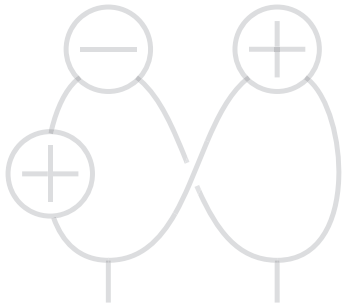
A *Hopf diagram* is a diagram with inputs but no output made up of the following lego pieces:



Examples:



a Hopf diagram
with 1 input

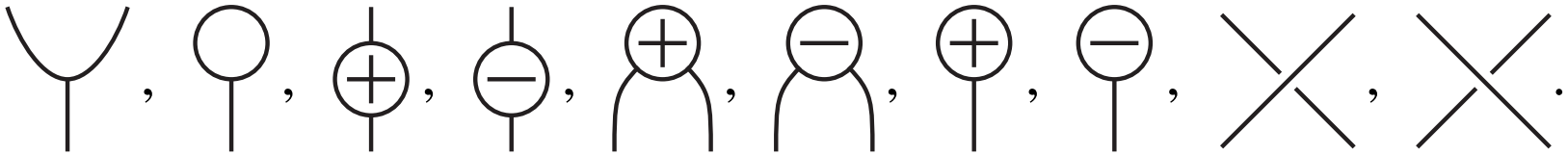


a Hopf diagram
with 2 inputs

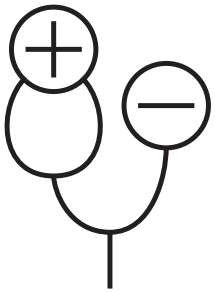
Put relations on Hopf diagrams $\rightsquigarrow$ monoidal category **HDiag**

Hopf diagrams

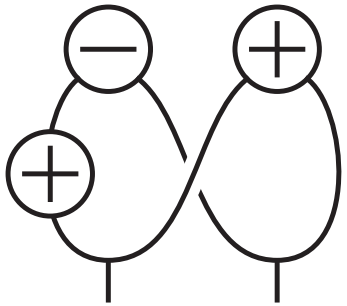
A *Hopf diagram* is a diagram with inputs but no output made up of the following lego pieces:



Examples:



a Hopf diagram
with 1 input

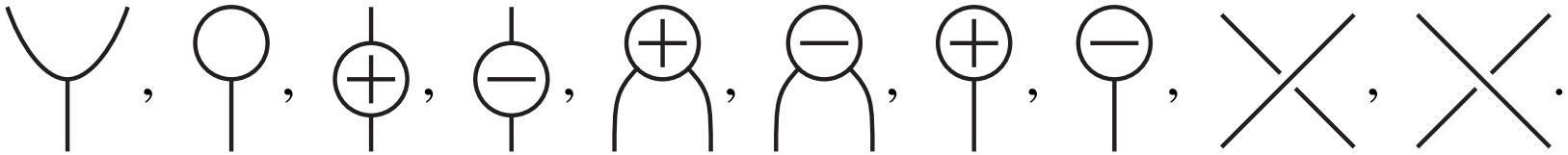


a Hopf diagram
with 2 inputs

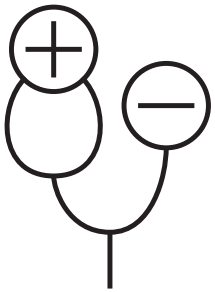
Put relations on Hopf diagrams $\rightsquigarrow$ monoidal category **HDiag**

Hopf diagrams

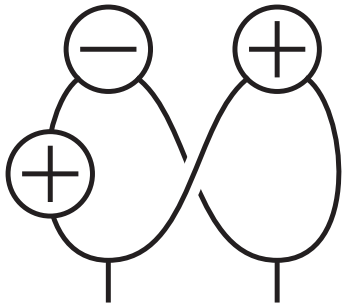
A *Hopf diagram* is a diagram with inputs but no output made up of the following lego pieces:



Examples:



a Hopf diagram
with 1 input

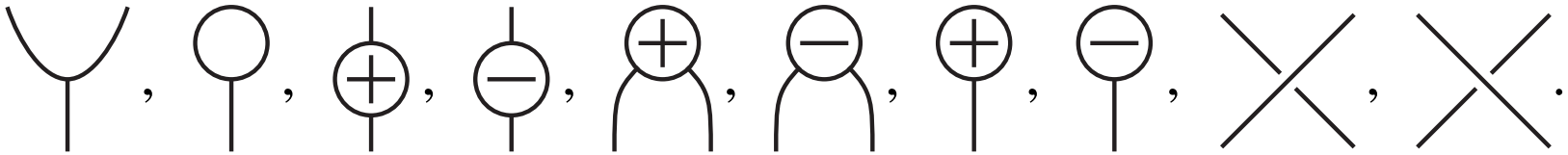


a Hopf diagram
with 2 inputs

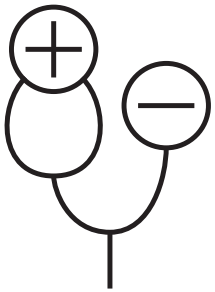
Put relations on Hopf diagrams $\rightsquigarrow$ monoidal category **HDiag**

Hopf diagrams

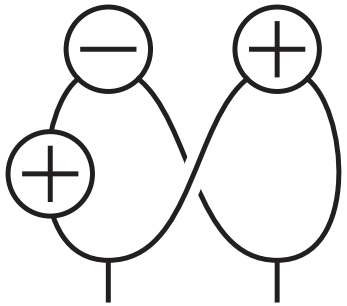
A *Hopf diagram* is a diagram with inputs but no output made up of the following lego pieces:



Examples:



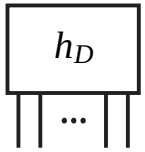
a Hopf diagram
with 1 input

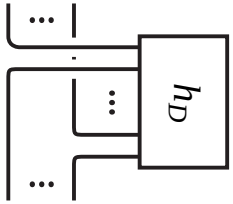


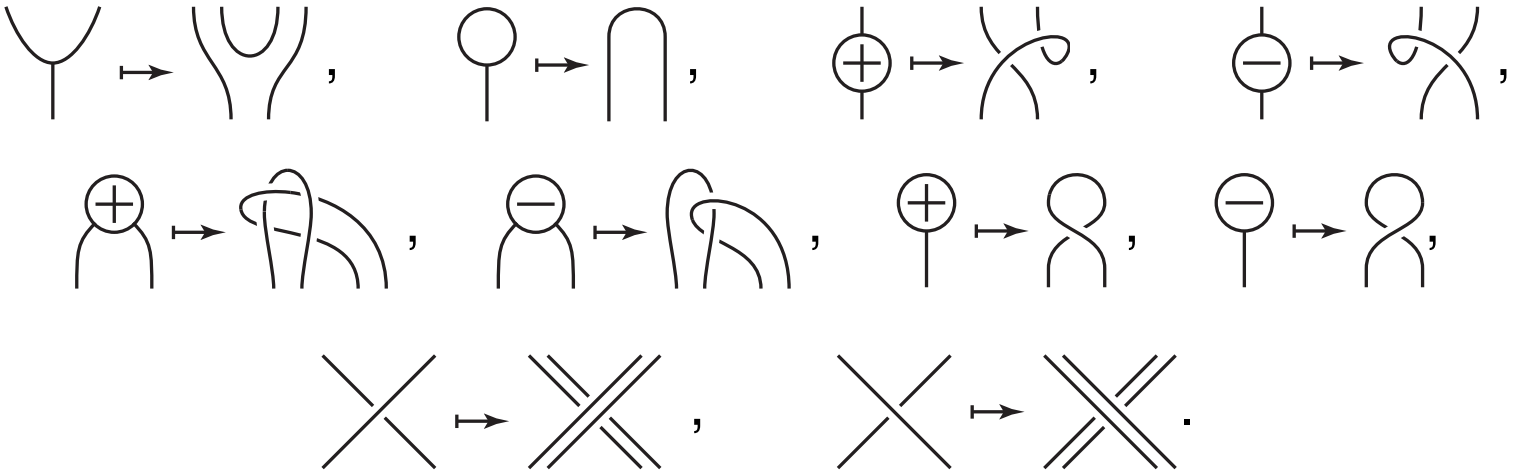
a Hopf diagram
with 2 inputs

Put relations on Hopf diagrams $\rightsquigarrow$ monoidal category **HDiag**

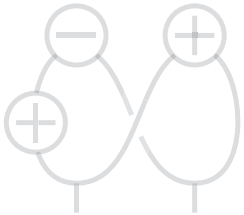
From Hopf diagrams to string links

D Hopf diagram $\mapsto$ 

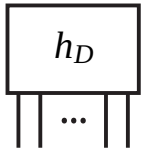
$\rightsquigarrow$ $\Phi(D) =$ 

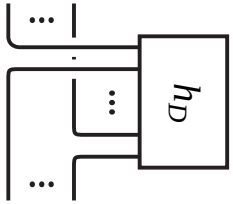


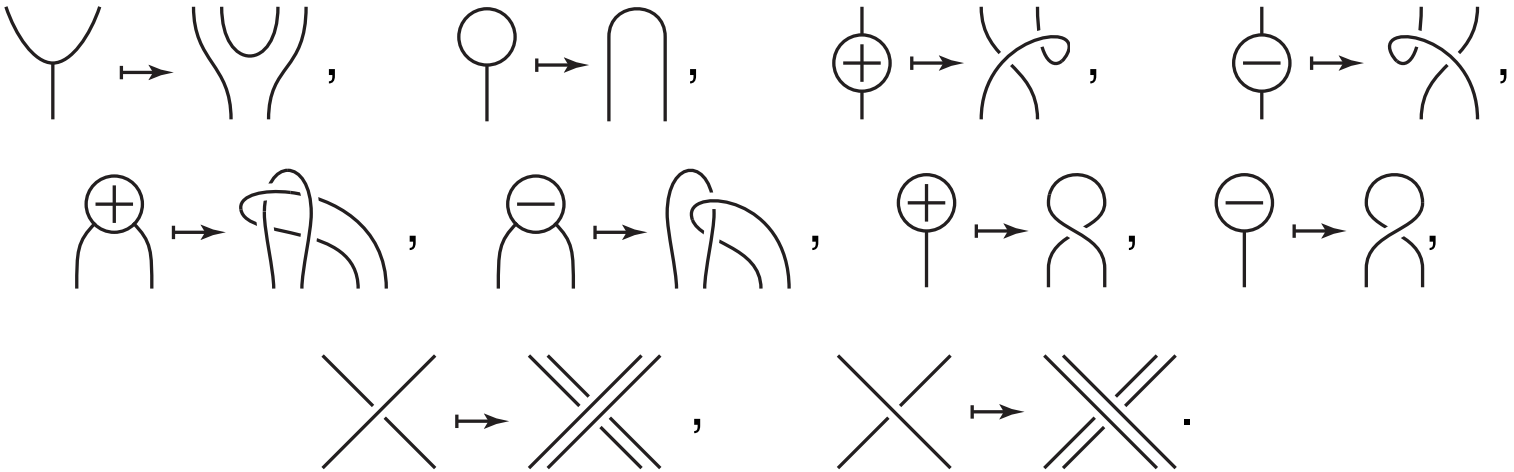
Example:

$D =$ 

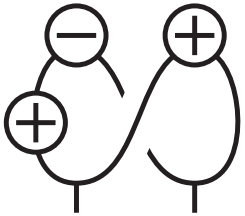
From Hopf diagrams to string links

D Hopf diagram $\mapsto$ 

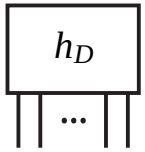
$\rightsquigarrow$ $\Phi(D) =$ 

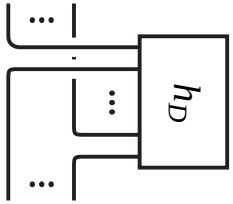


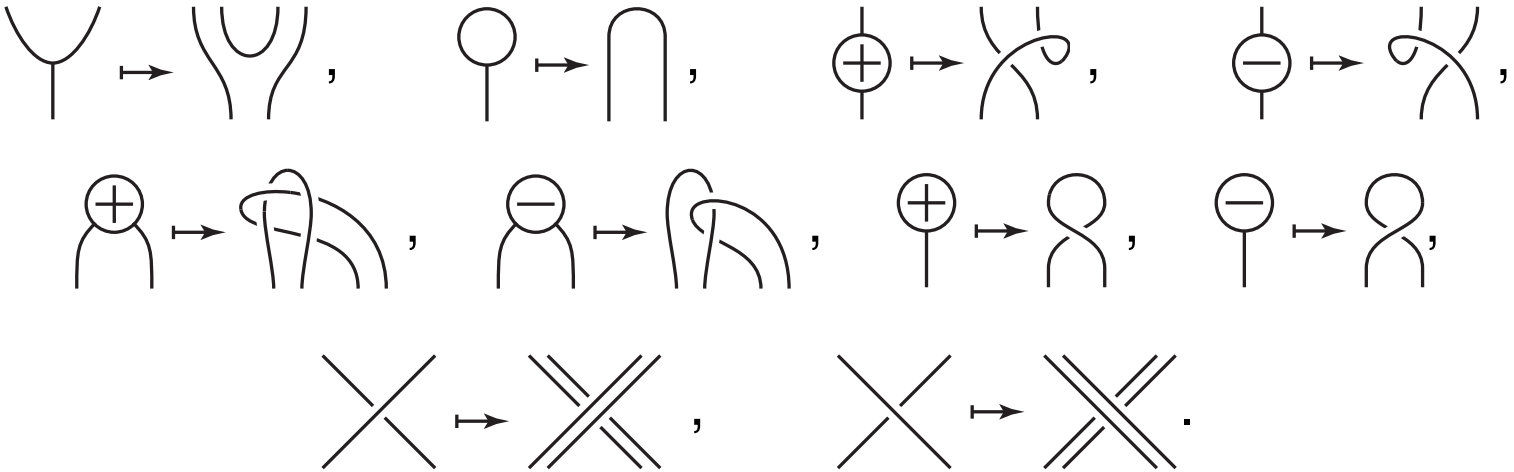
Example:

$D =$ 

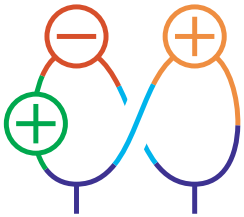
From Hopf diagrams to string links

D Hopf diagram $\mapsto$ 

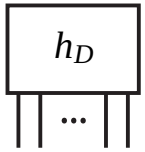
$\rightsquigarrow$ $\Phi(D) =$ 

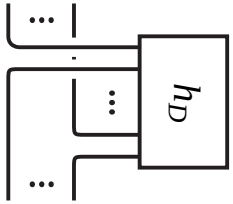


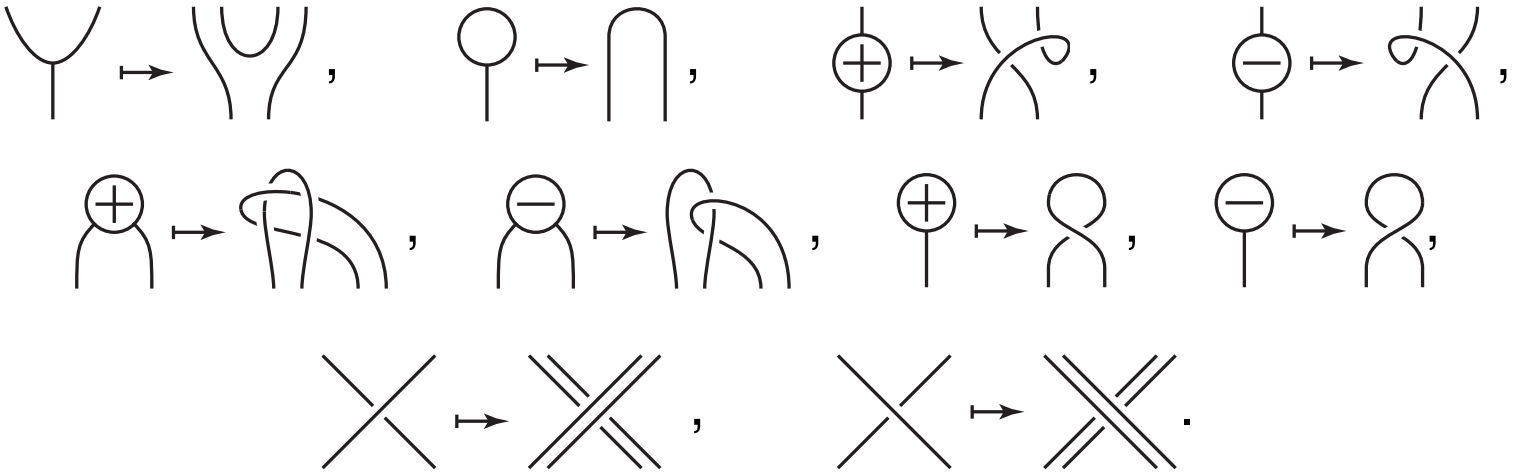
Example:

$D =$ 

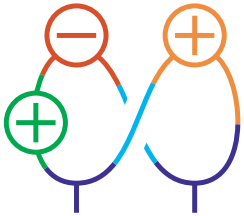
From Hopf diagrams to string links

D Hopf diagram $\mapsto$ 

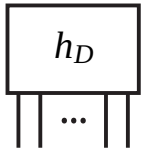
$\rightsquigarrow$ $\Phi(D) =$ 

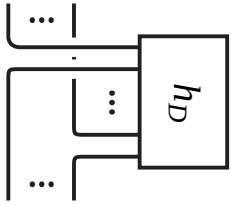


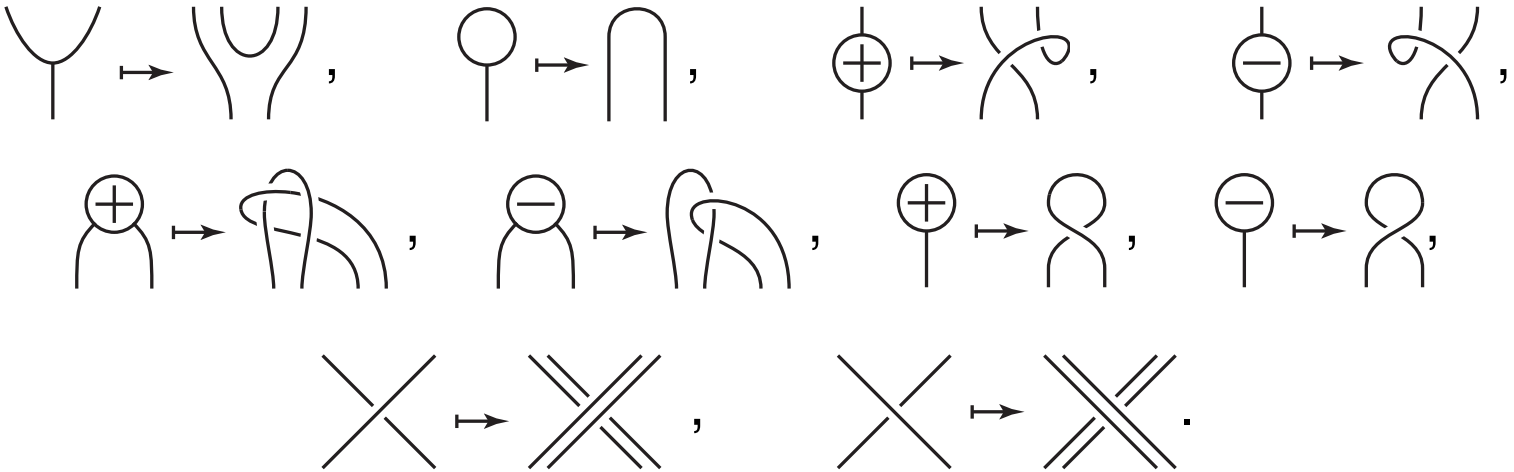
Example:

$D =$  $\mapsto$ $h_D =$

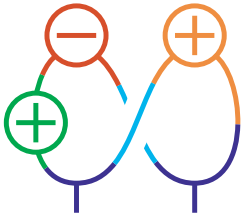
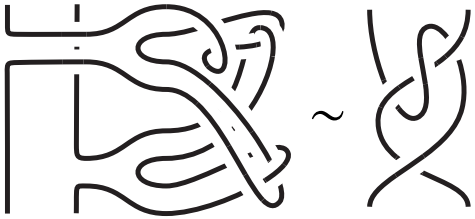
From Hopf diagrams to string links

D Hopf diagram $\mapsto$ 

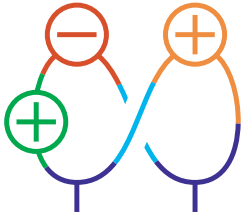
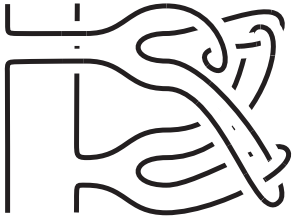
$\rightsquigarrow$ $\Phi(D) =$ 



Example:

$D =$  $\mapsto$ $h_D =$ $\rightsquigarrow$ $\Phi(D) =$ 

Hopf diagrams vs string links

$D =$  $\mapsto$ $\rightsquigarrow$ $\Phi(D) =$  $\sim$

Theorem (B.-V., 2005)

1. $\Phi\colon$ **HDiag** $\to$ **StLink** *is a well-defined monoidal functor*
2. *It is surjective*
3. *Better:* Φ *has a monoidal section* $\Psi\colon$ **StLink** $\to$ **HDiag**

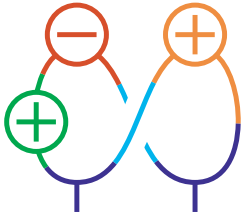
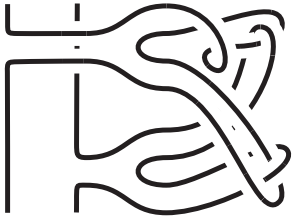
$$\Phi \circ \Psi = 1_{\mathbf{StLink}} \quad \textit{(constructive proof)}$$

Application: construct invariants by representing **HDiag**

Ex: $M_n(\mathbb{C}[q, q^{-1}])$, $\quad \Delta(E_{i,j}) = \sum_{k=1}^{n} E_{i,k} \otimes E_{k,j}$

gives rise to the n-colored Jones polynomial

Hopf diagrams vs string links

$D =$  $\mapsto$ $\rightsquigarrow$ $\Phi(D) =$  $\sim$

Theorem (B.-V., 2005)

1. $\Phi\colon$ **HDiag** $\to$ **StLink** *is a well-defined monoidal functor*
2. *It is surjective*
3. *Better:* Φ *has a monoidal section* $\Psi\colon$ **StLink** $\to$ **HDiag**

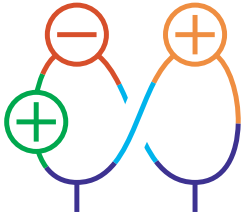
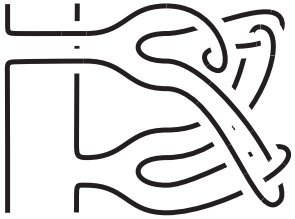
$$\Phi \circ \Psi = 1_{\mathbf{StLink}} \quad \textit{(constructive proof)}$$

Application: construct invariants by representing **HDiag**

Ex: $M_n(\mathbb{C}[q, q^{-1}])$, $\quad \Delta(E_{i,j}) = \sum_{k=1}^{n} E_{i,k} \otimes E_{k,j}$

gives rise to the n-colored Jones polynomial

Hopf diagrams vs string links

$D =$  $\mapsto$ $\rightsquigarrow$ $\Phi(D) =$  $\sim$

Theorem (B.-V., 2005)

1. $\Phi\colon$ **HDiag** $\to$ **StLink** *is a well-defined monoidal functor*
2. *It is surjective*
3. *Better:* Φ *has a monoidal section* $\Psi\colon$ **StLink** $\to$ **HDiag**

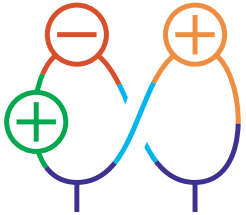
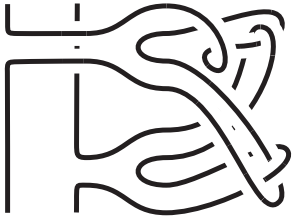
$$\Phi \circ \Psi = 1_{\mathbf{StLink}} \quad \textit{(constructive proof)}$$

Application: construct invariants by representing **HDiag**

Ex: $M_n(\mathbb{C}[q, q^{-1}])$, $\quad \Delta(E_{i,j}) = \sum_{k=1}^{n} E_{i,k} \otimes E_{k,j}$

gives rise to the n-colored Jones polynomial

Hopf diagrams vs string links

$D =$  $\mapsto$ $\rightsquigarrow$ $\Phi(D) =$  $\sim$

Theorem (B.-V., 2005)

1. $\Phi\colon$ **HDiag** $\to$ **StLink** *is a well-defined monoidal functor*
2. *It is surjective*
3. *Better:* Φ *has a monoidal section* $\Psi\colon$ **StLink** $\to$ **HDiag**

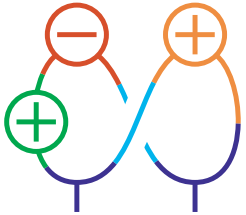
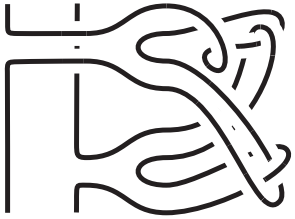
$$\Phi \circ \Psi = 1_{\mathbf{StLink}} \quad \textit{(constructive proof)}$$

Application: construct invariants by representing **HDiag**

Ex: $M_n(\mathbb{C}[q, q^{-1}])$, $\quad \Delta(E_{i,j}) = \sum_{k=1}^{n} E_{i,k} \otimes E_{k,j}$

gives rise to the n-colored Jones polynomial

Hopf diagrams vs string links

$D =$  $\mapsto$ $\rightsquigarrow$ $\Phi(D) =$  $\sim$

Theorem (B.-V., 2005)

1. $\Phi\colon$ **HDiag** $\to$ **StLink** *is a well-defined monoidal functor*
2. *It is surjective*
3. *Better:* Φ *has a monoidal section* $\Psi\colon$ **StLink** $\to$ **HDiag**

$$\Phi \circ \Psi = 1_{\mathbf{StLink}} \quad \textit{(constructive proof)}$$

Application: construct invariants by representing **HDiag**

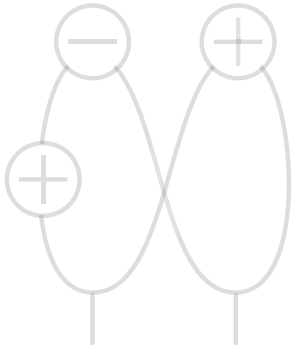
Ex: $M_n(\mathbb{C}[q, q^{-1}])$, $\quad \Delta(E_{i,j}) = \sum_{k=1}^{n} E_{i,k} \otimes E_{k,j}$

gives rise to the n-colored Jones polynomial

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ with $T =$ encoded by $D_T =$

$D_T =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

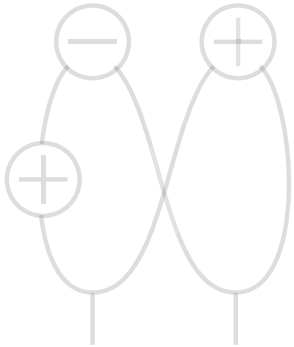
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ with $T =$ encoded by $D_T =$

$D_T =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

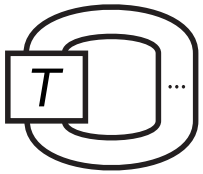
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

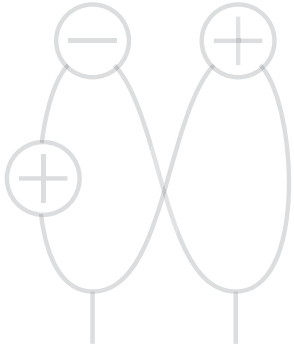
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$

$D_T =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

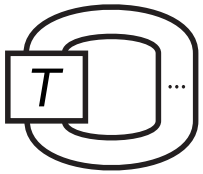
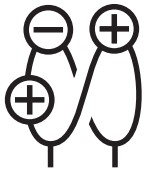
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

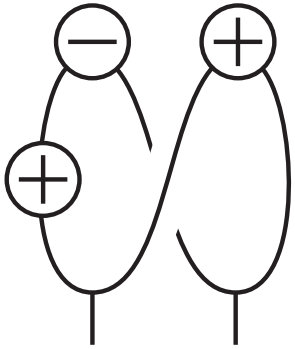
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$  with $T =$ encoded by $D_T =$ 

$D_T =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

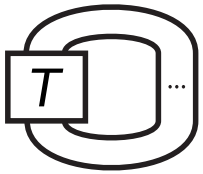
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

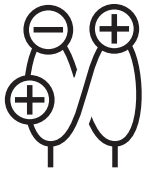
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

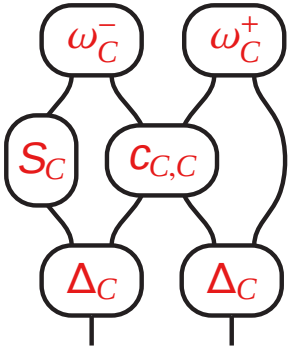
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$ 

$\phi_T =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

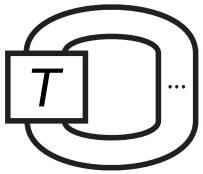
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

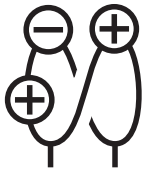
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

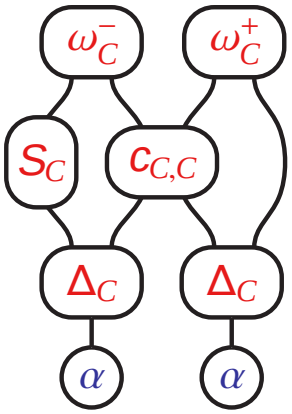
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$ 

$\tau_{\mathcal{B}}(M;\alpha) =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

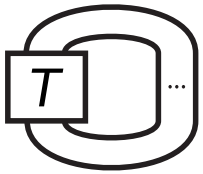
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

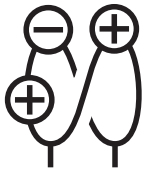
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

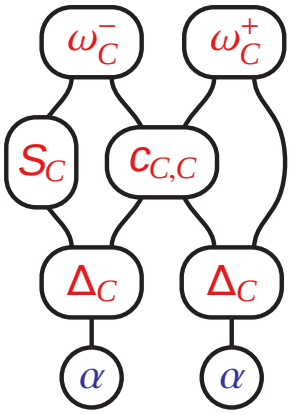
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$ 

$\tau_{\mathcal{B}}(M;\alpha) =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

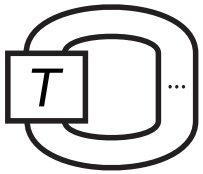
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

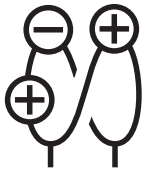
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

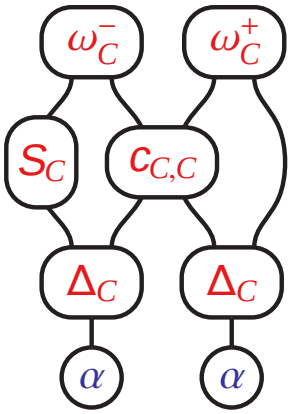
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$ 

$\tau_{\mathcal{B}}(M;\alpha) =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

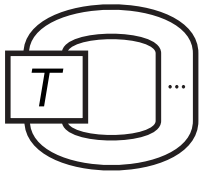
$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

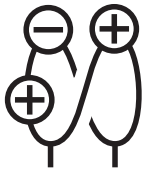
The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

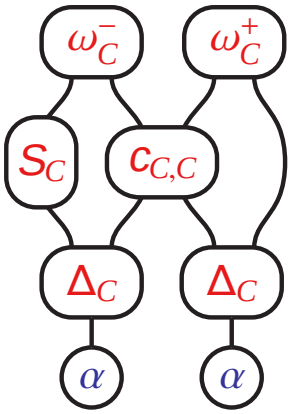
$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$
$= \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Summary

$\mathcal{B}$ braided ribbon category admitting a coend C

$M \simeq S^3_L$, $L \sim$ 

with $T =$ encoded by $D_T =$ 

$\tau_{\mathcal{B}}(M;\alpha) =$ 

structural morphisms of $C = \mathrm{coend}(\mathcal{B})$

Kirby element of $\mathcal{B}$

$\tau_{\mathcal{B}}(M;\alpha)$ depends only on:

The coproduct Δ_C, counit ε_C, antipode S_C, Hopf paring ω_C, twist θ_C, braiding $c_{C,C}$, and Kirby element α (ex: integral λ_C)

$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) = \tau_{\mathcal{Z}(\mathcal{C})}(M;\lambda_C)$ $\rightsquigarrow$ we need to describe the coend of $\mathcal{Z}(\mathcal{C})$ in terms of $\mathcal{C}$ $\rightsquigarrow$ *Hopf monad*

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{Hom}_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{Hom}_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A \otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a bialgebra $\Leftrightarrow \begin{cases} {}_A\mathrm{mod} \text{ is monoidal} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A\otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a bialgebra $\Leftrightarrow$ $\begin{cases} {}_A\mathrm{mod} \text{ is monoidal} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A\otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a bialgebra $\Leftrightarrow$ $\begin{cases} {}_A\mathrm{mod} \text{ is monoidal} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A\otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a bialgebra $\Leftrightarrow$ $\begin{cases} {}_A\mathrm{mod} \text{ is monoidal} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A\otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a Hopf algebra $\Leftrightarrow$ $\begin{cases} {}_A\mathrm{mod} \text{ is monoidal with duals} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F = A\otimes_{\Bbbk} ?"] \end{tikzcd}$$

Tannaka duality

A is a Hopf algebra $\Leftrightarrow$ $\begin{cases} {}_A\mathrm{mod} \text{ is monoidal with duals} \\ U \text{ strong monoidal} \end{cases}$

This characterization does not use the braiding on $\mathrm{vect}_{\Bbbk}$

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T : \mathcal{C} \to \mathcal{C}, \quad \mu : T^2 \to T \text{ (product)}, \quad \eta : 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U : T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y} : T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Towards Hopf algebras in the non-braided setting

A finite dim. algebra over $\Bbbk$ field. We have a natural bijection:

$$\mathrm{H}om_A(A \otimes_{\Bbbk} X, Y) \simeq \mathrm{H}om_{\Bbbk}(X, Y)$$

In other words we have an adjunction:

$$\begin{tikzcd} {}_A\mathrm{mod} \arrow[d, bend left, "U"] \\ \mathrm{vect}_{\Bbbk} \arrow[u, bend left, "F=A\otimes_{\Bbbk}?"] \end{tikzcd}$$

$$T = UF\colon \left\{ \begin{array}{ccc} \mathrm{vect}_{\Bbbk} & \to & \mathrm{vect}_{\Bbbk} \\ X & \mapsto & A \otimes_k X \end{array} \right. \text{ is a monad}$$

T- $\mathrm{vect}_{\Bbbk} \simeq {}_A\mathrm{mod}$

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$\Leftrightarrow$ T comonoidal: $\Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Monads and bimonads

- $\mathcal{C}$ category $\rightsquigarrow \mathrm{EndoFun}(\mathcal{C})$ monoidal category ($\otimes = \circ$, $\mathbb{1} = 1_{\mathcal{C}}$)

A monad on $\mathcal{C}$ is a monoid in $\mathrm{EndoFun}(\mathcal{C})$:

$$T\colon \mathcal{C} \to \mathcal{C}, \quad \mu\colon T^2 \to T \text{ (product)}, \quad \eta\colon 1_{\mathcal{C}} \to T \text{ (unit)}$$

T-$\mathcal{C}$= category of *T-modules* in $\mathcal{C}$

$U\colon T\text{-}\mathcal{C} \to \mathcal{C}$ forgetful functor $\rightsquigarrow$ $\begin{tikzcd} T\text{-}\mathcal{C} \arrow[r, bend left, "U"] & \mathcal{C} \arrow[l, bend left, "F"] \end{tikzcd}$ adjunction

- $\mathcal{C}$ monoidal category

Moerdijk (02) : T bimonad if $\begin{cases} T\text{-}\mathcal{C} \text{ monoidal} \\ U \text{ strong monoidal} \end{cases}$

$$\Leftrightarrow T \text{ comonoidal: } \Delta_{X,Y}\colon T(X \otimes Y) \to T(X) \otimes T(Y)$$

Example: (F, U) monoidal adjunction $\Rightarrow T = UF$ bimonad

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow$ $T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

$\mathcal{C}$ monoidal category, (T, μ, η) bimonad on $\mathcal{C}$, $U\colon T\text{-}\mathcal{C} \to \mathcal{C}$

T Hopf monad if its *fusion operators*

$$F_{X,Y} = (\mathrm{id}_{T(X)} \otimes \mu_Y)\Delta_{X,T(Y)}\colon T(X \otimes T(Y)) \to T(X) \otimes T(Y)$$

are isomorphisms

Theorem (B.V. 2007, B.L.V. 2009)

When $\mathcal{C}$ has duals (or internal homs):

T Hopf monad $\Leftrightarrow$ $T\text{-}\mathcal{C}$ has duals (or internal homs) preserved by U

$\Leftrightarrow$ T has an antipode

Example: $\mathcal{D} \underset{F}{\overset{U}{\rightleftarrows}} \mathcal{C}$ monoidal adjunction between rigid categories

$\rightsquigarrow T = UF$ is a Hopf monad on $\mathcal{C}$

Hopf monads

Example: A Hopf algebra in braided category $\mathcal{B}$

$$\rightsquigarrow \quad T(X) = A \otimes X \text{ is a Hopf monad on } \mathcal{B}$$

Theorem (B.L.V., 2009)

Characterization of the representable Hopf monads

Outlook of General Theory of Hopf monads

- Hopf modules and Sweedler decomposition theorem
- Existence of universal integrals
  (generalizes Bespalov-Kerler-Lyubashenko-Turaev)
- Semisimplicity, Maschke criterion
- Cross-products, bosonization for Hopf monads
- Quasitriangular Hopf monads, double

Hopf monads

Example: A Hopf algebra in braided category $\mathcal{B}$

$$\rightsquigarrow \quad T(X) = A \otimes X \text{ is a Hopf monad on } \mathcal{B}$$

Theorem (B.L.V., 2009)

Characterization of the representable Hopf monads

Outlook of General Theory of Hopf monads

- Hopf modules and Sweedler decomposition theorem
- Existence of universal integrals
  (generalizes Bespalov-Kerler-Lyubashenko-Turaev)
- Semisimplicity, Maschke criterion
- Cross-products, bosonization for Hopf monads
- Quasitriangular Hopf monads, double

Hopf monads

Example: A Hopf algebra in braided category $\mathcal{B}$

$$\rightsquigarrow \quad T(X) = A \otimes X \text{ is a Hopf monad on } \mathcal{B}$$

Theorem (B.L.V., 2009)

Characterization of the representable Hopf monads

Outlook of General Theory of Hopf monads

- Hopf modules and Sweedler decomposition theorem
- Existence of universal integrals
  (generalizes Bespalov-Kerler-Lyubashenko-Turaev)
- Semisimplicity, Maschke criterion
- Cross-products, bosonization for Hopf monads
- Quasitriangular Hopf monads, double

Hopf monads as ‘quantum groupoids’

R ring $\rightsquigarrow$ $({}_R\mathrm{Mod}_R, {}_R\otimes_R, R)$ monoidal category

Facts

- Linear bimonads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *bialgebroids* in the sense of Takeuchi
- Linear Hopf monads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *Hopf algebroids* in the sense of Schauenburg

The axiomatic of Hopf monads is easier to manipulate

Theorem (B.L.V., 2009)

A finite tensor category over a field is tensor equivalent to the category of finite-dimensional modules over some Hopf algebroid

Hopf monads as 'quantum groupoids'

R ring $\rightsquigarrow$ $({}_R\mathrm{Mod}_R, {}_R\otimes_R, R)$ monoidal category

Facts

- Linear bimonads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *bialgebroids* in the sense of Takeuchi
- Linear Hopf monads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *Hopf algebroids* in the sense of Schauenburg

The axiomatic of Hopf monads is easier to manipulate

Theorem (B.L.V., 2009)

A finite tensor category over a field is tensor equivalent to the category of finite-dimensional modules over some Hopf algebroid

Hopf monads as 'quantum groupoids'

R ring $\rightsquigarrow$ $({}_R\mathrm{Mod}_R, {}_R\otimes_R, R)$ monoidal category

Facts

- Linear bimonads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *bialgebroids* in the sense of Takeuchi
- Linear Hopf monads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *Hopf algebroids* in the sense of Schauenburg

The axiomatic of Hopf monads is easier to manipulate

Theorem (B.L.V., 2009)

A finite tensor category over a field is tensor equivalent to the category of finite-dimensional modules over some Hopf algebroid

Hopf monads as 'quantum groupoids'

R ring $\rightsquigarrow$ $({}_R\mathrm{Mod}_R, {}_R\otimes_R, R)$ monoidal category

Facts

- Linear bimonads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *bialgebroids* in the sense of Takeuchi
- Linear Hopf monads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *Hopf algebroids* in the sense of Schauenburg

The axiomatic of Hopf monads is easier to manipulate

Theorem (B.L.V., 2009)

A finite tensor category over a field is tensor equivalent to the category of finite-dimensional modules over some Hopf algebroid

Hopf monads as 'quantum groupoids'

R ring $\rightsquigarrow$ $({}_R\mathrm{Mod}_R, {}_R\otimes_R, R)$ monoidal category

Facts

- Linear bimonads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *bialgebroids* in the sense of Takeuchi
- Linear Hopf monads on ${}_R\mathrm{Mod}_R$ with right adjoints are exactly *Hopf algebroids* in the sense of Schauenburg

The axiomatic of Hopf monads is easier to manipulate

Theorem (B.L.V., 2009)

A finite tensor category over a field is tensor equivalent to the category of finite-dimensional modules over some Hopf algebroid

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

Centralizer and double of a Hopf monad

T Hopf monad on $\mathcal{C}$ rigid, which is centralisable:

$$Z_T(X) = \int^{Y \in \mathcal{C}} T(Y)^* \otimes X \otimes Y \quad \text{exists for all } X \in \mathrm{Ob}(X)$$

Then $Z_T : \mathcal{C} \to \mathcal{C}$ is an endofunctor of $\mathcal{C}$ (the centralizer of T)

Theorem (B.V., 2008)

1. Z_T is a Hopf monad on $\mathcal{C}$
2. There is a canonical distributive law $\Omega : T \circ Z_T \to Z_T \circ T$
3. $D_T = Z_T \circ_\Omega T$ is a quasitriangular Hopf monad on $\mathcal{C}$
4. $\mathcal{Z}(T\text{-}\mathcal{C}) \cong D_T\text{-}\mathcal{C}$ as braided categories

Applications :
- $\mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$ where $Z = Z_{1_\mathcal{C}} = D_{1_\mathcal{C}}$
- $\mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1})$ has an explicit description

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, $I =$ rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2008)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modulaire (i.e. $\omega_{\mathcal{C}} \colon C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, $I =$ rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2008)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modulaire (i.e. $\omega_{\mathcal{C}} \colon C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, $I =$ rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2008)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modulaire (i.e. $\omega_{\mathcal{C}} \colon C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, $I =$ rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2008)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modulaire (i.e. $\omega_{\mathcal{C}} : C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, I = rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2008)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

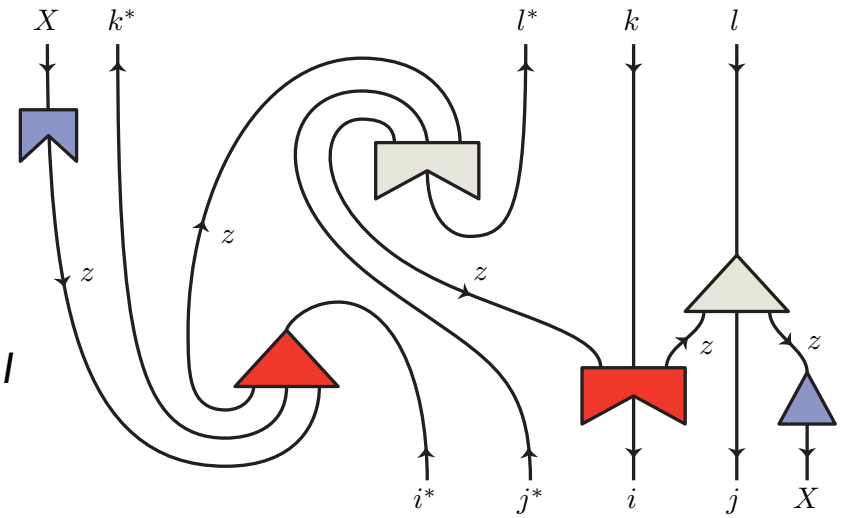
Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modulaire (i.e. $\omega_{\mathcal{C}} : C \otimes C \to \mathbb{1}$ non-degenerate)

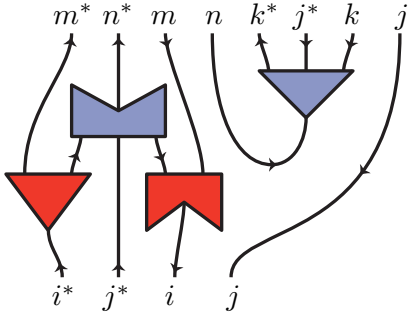
  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

Structural Morphisms of the coend of $\mathcal{Z}(\mathcal{C})$

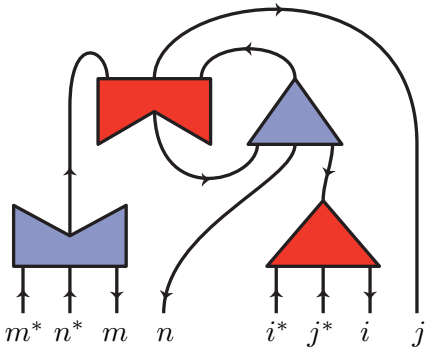
$$C = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j \qquad \sigma_X = \sum_{i,j,k,l,z \in I}$$



$$\Delta_C = \sum_{i,j,k,m,n \in I}$$



$$\omega_C = \sum_{i,j,k,l \in I}$$



The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, I = rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \operatorname{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2009)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modular (i.e. $\omega_{\mathcal{C}} : C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, I = rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \mathrm{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2009)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modular (i.e. $\omega_{\mathcal{C}} : C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, I = rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \qquad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \operatorname{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2009)

1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

Applications:

- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modular (i.e. $\omega_C \colon C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

The center of a fusion category

$\mathcal{C}$ spherical fusion over a **ring** $\Bbbk$, $I =$ rep. set of simples of $\mathcal{C}$

$$Z(X) = Z_{1_{\mathcal{C}}}(X) = \bigoplus_{i \in I} i^* \otimes X \otimes i, \quad \mathcal{Z}(\mathcal{C}) \simeq Z\text{-}\mathcal{C}$$

$$C = \operatorname{coend}\big(Z(\mathcal{C})\big) = Z_Z(\mathbb{1}) = \bigoplus_{i,j \in I} i^* \otimes j^* \otimes i \otimes j$$

Théorème (B.V., 2009)

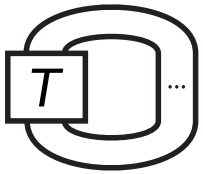
1. Z semisimple $\Leftrightarrow$ $\dim \mathcal{C}$ invertible in $\Bbbk$
2. $\lambda = \sum_{j \in I} \dim_q(j)$ $: \mathbb{1} \to C$ is an integral of C

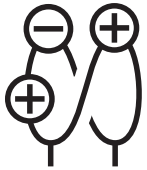
Applications:

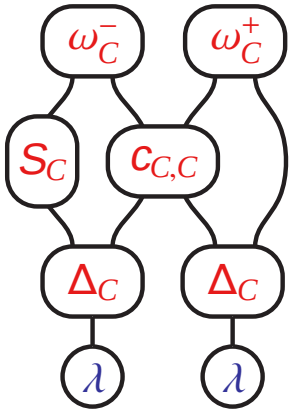
- $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0 \;\Rightarrow\; \mathcal{Z}(\mathcal{C})$ fusion
- $\mathcal{Z}(\mathcal{C})$ is modular (i.e. $\omega_C \colon C \otimes C \to \mathbb{1}$ non-degenerate)

  $\rightsquigarrow$ Müger's thm when $\Bbbk$ alg. clos field and $\dim \mathcal{C} \neq 0$

Computation of $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$

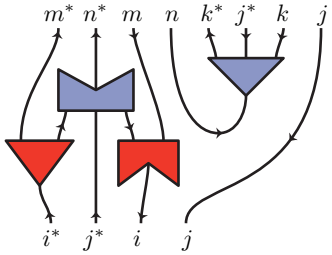
$M \simeq S^3_L \quad L \sim$ 

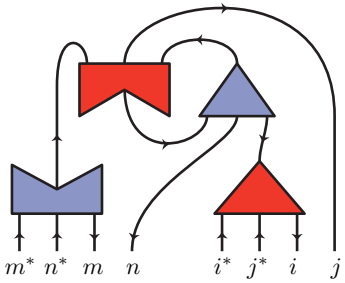
avec $T =$ $\leftsquigarrow$ 

$\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M) =$ 

morphismes de structure de C

intégrale de C

$\Delta_C = \sum_{i,j,k,m,n \in I}$ 

$\omega_C = \sum_{i,j,k,l \in I}$ 

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim \mathcal{C} \neq 0$

Theorem (T.V., 2010)

1. $\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ for all 3-manifold M
2. $\mathrm{TV}_{\mathcal{C}}$ et $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}$ are isomorphic TQFT

Key points of the proof:

- Extend $\mathrm{TV}_{\mathcal{C}}$ to a TQFT for 3-cobordims endowed with $\mathcal{Z}(\mathcal{C})$-colored framed (using skeletons)
- For any genus g surface Σ,

$$\mathrm{TV}_{\mathcal{C}}(\Sigma) \simeq \mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1}, C^{\otimes g}) \simeq \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(\Sigma)$$

where C= coend of $\mathcal{Z}(\mathcal{C})$

Applications :
- computation of $\mathrm{TV}_{\mathcal{C}}(M)$ via Hopf diagrams
- $\mathcal{C}$ unitary $\Rightarrow \mathrm{TV}_{\mathcal{C}}$ is a unitary TQFT

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim \mathcal{C} \neq 0$

Theorem (T.V., 2010)

1. $\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ for all 3-manifold M
2. $\mathrm{TV}_{\mathcal{C}}$ et $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}$ are isomorphic TQFT

Key points of the proof:

- Extend $\mathrm{TV}_{\mathcal{C}}$ to a TQFT for 3-cobordims endowed with $\mathcal{Z}(\mathcal{C})$-colored framed (using skeletons)
- For any genus g surface Σ,

$$\mathrm{TV}_{\mathcal{C}}(\Sigma) \simeq \mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1}, C^{\otimes g}) \simeq \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(\Sigma)$$

where C= coend of $\mathcal{Z}(\mathcal{C})$

Applications : • computation of $\mathrm{TV}_{\mathcal{C}}(M)$ via Hopf diagrams

• $\mathcal{C}$ unitary $\Rightarrow$ $\mathrm{TV}_{\mathcal{C}}$ is a unitary TQFT

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim \mathcal{C} \neq 0$

Theorem (T.V., 2010)

1. $\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ for all 3-manifold M
2. $\mathrm{TV}_{\mathcal{C}}$ et $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}$ are isomorphic TQFT

Key points of the proof:

- Extend $\mathrm{TV}_{\mathcal{C}}$ to a TQFT for 3-cobordims endowed with $\mathcal{Z}(\mathcal{C})$-colored framed (using skeletons)
- For any genus g surface Σ,

$$\mathrm{TV}_{\mathcal{C}}(\Sigma) \simeq \mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1}, C^{\otimes g}) \simeq \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(\Sigma)$$

where C= coend of $\mathcal{Z}(\mathcal{C})$

Applications : • computation of $\mathrm{TV}_{\mathcal{C}}(M)$ via Hopf diagrams

• $\mathcal{C}$ unitary $\Rightarrow$ $\mathrm{TV}_{\mathcal{C}}$ is a unitary TQFT

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim \mathcal{C} \neq 0$

Theorem (T.V., 2010)

1. $\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ for all 3-manifold M
2. $\mathrm{TV}_{\mathcal{C}}$ et $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}$ are isomorphic TQFT

Key points of the proof:

- Extend $\mathrm{TV}_{\mathcal{C}}$ to a TQFT for 3-cobordims endowed with $\mathcal{Z}(\mathcal{C})$-colored framed (using skeletons)
- For any genus g surface Σ,

$$\mathrm{TV}_{\mathcal{C}}(\Sigma) \simeq \mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1}, C^{\otimes g}) \simeq \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(\Sigma)$$

where C= coend of $\mathcal{Z}(\mathcal{C})$

Applications : • computation of $\mathrm{TV}_{\mathcal{C}}(M)$ via Hopf diagrams

• $\mathcal{C}$ unitary $\Rightarrow$ $\mathrm{TV}_{\mathcal{C}}$ is a unitary TQFT

Comparing RT and TV

$\mathcal{C}$ spherical fusion category with $\dim \mathcal{C} \neq 0$

Theorem (T.V., 2010)

1. $\mathrm{TV}_{\mathcal{C}}(M) = \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(M)$ for all 3-manifold M
2. $\mathrm{TV}_{\mathcal{C}}$ et $\mathrm{RT}_{\mathcal{Z}(\mathcal{C})}$ are isomorphic TQFT

Key points of the proof:

- Extend $\mathrm{TV}_{\mathcal{C}}$ to a TQFT for 3-cobordims endowed with $\mathcal{Z}(\mathcal{C})$-colored framed (using skeletons)
- For any genus g surface Σ,

$$\mathrm{TV}_{\mathcal{C}}(\Sigma) \simeq \mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}(\mathbb{1}, C^{\otimes g}) \simeq \mathrm{RT}_{\mathcal{Z}(\mathcal{C})}(\Sigma)$$

where C= coend of $\mathcal{Z}(\mathcal{C})$

Applications :
- computation of $\mathrm{TV}_{\mathcal{C}}(M)$ via Hopf diagrams
- $\mathcal{C}$ unitary $\Rightarrow \mathrm{TV}_{\mathcal{C}}$ is a unitary TQFT